

Block-Separable Copula Generated Random Graphs for Directed Networks

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Abstract

We propose a model for directed, real-valued networks in which the dependence between the two directed edges within each dyad is captured by a one-parameter Frank copula selected on the basis of empirical dependence structure and predictive likelihood, while the marginal mean of each edge is decomposed into additive sender and receiver effects. The model is closed-form, tractable under componentwise Metropolis–Hastings sampling, and designed for replicated networks (multiple networks observed on a common node set, each belonging to one of G comparison groups).

We derive several theoretical properties that clarify the interpretation and estimability of the dyadic reciprocity parameter. The copula parameter ρ admits a closed-form, margin-invariant interpretation as a monotone function of Kendall’s τ via the Debye function. Under a standard normalisation of the sender effects, the full parameter vector is identifiable whenever $N \geq 3$. With fixed N and growing number of replicated networks S , the MLE of ρ is $\sqrt{S}$ -consistent and asymptotically normal under a simplified model serving as a parametric benchmark.

The model is applied to effective connectivity networks in the default mode network from resting-state fMRI for 112 participants across three diagnostic groups: Alzheimer’s disease, mild cognitive impairment, and cognitively normal. Posterior estimates of ρ are

strongly negative for all three groups ($\rho^{(g)} \approx -3.7$ to -3.8 , 95% credible intervals well below zero), indicating that within each dyad, the two directed edges are negatively associated: participants with a stronger effective connection A_{ij} tend to have a weaker reverse connection A_{ji} , consistent with a hierarchical regulation structure in the DMN. The three groups have nearly identical posterior distributions for ρ ; the posterior probability of the strict ordering $\rho^{(\text{NL})} < \rho^{(\text{MCI})} < \rho^{(\text{AD})}$ is 0.20, close to the uniform chance level of 1/6.

Keywords: Copulas, Directed networks, Reciprocity, Brain connectivity, Replicated networks, Alzheimer’s disease, Bayesian inference.

1 Introduction

Directed weighted networks, in which each ordered pair of nodes carries a real-valued edge weight, arise in neuroscience, economics, and the social sciences. In neuroscience, dynamic causal modelling (DCM; 14, 15) converts resting-state fMRI time series into an $N \times N$ matrix of effective connectivity strengths, one matrix per participant. The analyst then faces the problem of comparing the distributional properties of these connectivity matrices across diagnostic groups, such as Alzheimer’s disease (AD), mild cognitive impairment (MCI), and cognitively normal (NL) individuals [16, 18, 19].

A natural feature of such networks is *reciprocity*: the extent to which a strong directed edge A_{ij} tends to co-occur with a strong (or weak) reverse edge A_{ji} . Existing frameworks handle reciprocity in several ways. The p_1 model [30] and its extensions parametrize reciprocity via an explicit differential parameter. The additive and multiplicative effects (AME) model [9] captures reciprocity implicitly through multiplicative latent factors. The recent model of Feng and Leng [11] establishes consistency results for a directed network model with explicit sender/receiver effects and a reciprocity term. For real-valued edges specifically, Zhang et al. [21] develop a mixed-effects time-varying network model, and Lee and Kim [17] apply copula-based directional dependence to whole-brain connectivity from fMRI.

Unlike most network models that study a single large network growing in its number of nodes, our setting consists of many replicated networks on a small fixed node set. This regime arises naturally in neuroscience, where each participant contributes one network over a fixed set of brain regions, and motivates an asymptotic theory in which the number of replicates S grows while N is fixed.

This paper proposes a model for *replicated* directed real-valued networks that makes reciprocity both structurally explicit and statistically interpretable. The key idea is to model the two directed edges of each *dyad* (the unordered pair $\{i, j\}$) jointly via a bivariate Frank copula, while specifying the marginal distribution of each edge through additive sender and receiver node effects. Because dyads are treated as conditionally independent of one another,

the joint likelihood factorises over dyads, and the Frank copula parameter ρ becomes the sole driver of within-dyad dependence. We call this model a *Copula Generated Random Graph* (CGRG).

Our contributions are:

1. A parsimonious model for directed real-valued networks combining an interpretable additive mean structure with a single reciprocity parameter (Section 3).
2. A Bayesian hierarchical extension that pools information across G comparison groups through shared hyperparameters on the sender effects, receiver effects, and reciprocity parameters (Section 5).
3. Several theoretical results (Section 4): a closed-form, margin-invariant Kendall's τ formula, identifiability for $N \geq 3$, and a parametric benchmark establishing $\sqrt{S}$ -consistency of the MLE under fixed N and growing replicates S . (Appendix A provides an entropy decomposition that further clarifies the role of ρ but is not needed for the empirical results.)
4. An application to $N = 4$ default mode network connectivity matrices from 112 fMRI participants across AD, MCI, and NL groups (Section 8). The model recovers strongly negative reciprocity ($\rho \approx -3.75$, Kendall's $\tau \approx -0.43$) for all three groups with 95% credible intervals entirely below zero, indicating a hierarchical (non-reciprocal) regulatory structure in the DMN that is consistent across disease states. Posterior predictive checks confirm that the model reproduces the observed dyadic Kendall's τ pattern.

Section 2 introduces the dyad-separable framework. Section 3 specifies the Gaussian-marginal Frank-copula model. Section 4 contains the core theoretical results (Kendall's τ , identifiability, and asymptotic consistency). Section 5 develops the hierarchical multi-group model for brain networks. Section 6 compares the CGRG to a multivariate normal baseline.

Section 7 presents simulation evidence on parameter recovery, including a null-signal scenario. Section 8 contains the brain application. Section 9 discusses extensions and concludes. Appendix A contains the entropy decomposition; Appendix B contains posterior density plots for node effects.

2 Dyad-Separable Copula Generated Random Graphs

Let G be a directed network on N nodes with adjacency matrix A . The key structural assumption of a copula generated random graph is that the $N(N-1)/2$ *dyads* (unordered pairs $\{i, j\}$, each comprising the two directed edges A_{ij} and A_{ji}) are conditionally independent of one another. Within each dyad, the joint distribution of (A_{ij}, A_{ji}) is specified via a copula binding their marginal CDFs. Formally, for any $\{i, j\}$ and $\{k, l\}$ with $\{i, j\} \neq \{k, l\}$,

$$(A_{ij}, A_{ji}) \perp (A_{kl}, A_{lk}) \mid \theta, \rho,$$

so the log-likelihood factorises as

$$\log p(A \mid \theta, \rho) = \sum_{i=1}^N \log p(A_{ii} \mid \theta) + \sum_{i < j} \log p(A_{ij}, A_{ji} \mid \theta, \rho). \quad (1)$$

This dyadic separability plays the same structural role that block-diagonal dependence plays in stochastic block models: it makes inference tractable and gives each marginal and copula parameter a direct interpretation. The parameter ρ governs *reciprocity*, the tendency of A_{ij} and A_{ji} to be concordant or discordant, and is common to all dyads within a group.

The assumption restricts dependence to occur only within unordered pairs of nodes. It excludes higher-order dependencies such as transitivity, triadic closure, or correlated sender and receiver effects across distinct dyads. The dyadic independence assumption is not intended to capture all forms of network dependence; rather, it isolates reciprocity as the dominant dependence mechanism while keeping inference tractable in replicated-network

104 settings. The assumption is deliberate: it yields a likelihood that factorises over dyads,
 105 reducing computation to a sum over $N(N-1)/2$ independent terms and isolating reciprocity
 106 as the primary dependence mechanism. In the replicated-network regime with small fixed N
 107 (the setting of the brain application in Section 5), dyadic separability provides a parsimonious
 108 alternative to high-dimensional covariance models that become inestimable when the number
 109 of replicates is modest relative to N^4 .

110 **3 A Dyadic Copula Model for Real-Valued Networks**

111 Many networks arising in the natural and social sciences carry real-valued edge weights. Reg-
 112 ulatory strength between brain regions, trade volumes between countries, and communication
 113 frequencies between colleagues are all examples. This section develops a CGRG model for
 114 directed networks with real-valued edges within the dyad-separable framework of Section 2.

115 **3.1 Model Specification**

116 Let G be a directed network on N nodes with edge weight matrix $A \in \mathbb{R}^{N \times N}$. Following the
 117 block-separable construction of Section 2, we partition the off-diagonal edges into M groups
 118 such that edges belonging to different groups are conditionally independent. We take the
 119 dyadic partition, so $M = N(N-1)/2$ and each group contains the two directed edges A_{ij} and
 120 A_{ji} connecting the same pair of nodes. Under this partition the log-likelihood decomposes as

$$\log p(A \mid \theta, \rho) = \sum_{i=1}^N \log p(A_{ii} \mid \theta) + \sum_{i=1}^{N-1} \sum_{j>i} \log p(A_{ij}, A_{ji} \mid \theta, \rho). \quad (2)$$

121 **3.2 Marginal Distributions and Identifiability**

122 The marginal distribution of each off-diagonal edge is

$$A_{ij} \mid \delta_i, \gamma_j, \lambda \sim \mathcal{N}(\delta_i + \gamma_j, \lambda^{-1}), \quad (3)$$

where δ_i is a *sender effect* measuring node i 's tendency to exert influence, γ_j is a *receiver effect* measuring node j 's tendency to be influenced, and $\lambda > 0$ is a precision parameter shared across edges [9].

The additive mean structure $\delta_i + \gamma_j$ is not identified without a constraint. We set $\delta_1 = 0$, so all sender effects are interpreted relative to node 1. Under this normalisation, γ_j equals the expected weight of edges directed into node j from node 1, and δ_i measures the excess sending tendency of node $i \geq 2$ relative to the reference.

3.3 Frank Copula Density for a Dyad

By Sklar's theorem [2], the joint distribution of (A_{ij}, A_{ji}) is expressed via a copula binding their marginal CDFs:

$$p(A_{ij}, A_{ji} \mid \theta, \rho) = c_\rho(F_{ij}(A_{ij}), F_{ji}(A_{ji})) \cdot f_{ij}(A_{ij}) \cdot f_{ji}(A_{ji}), \quad (4)$$

where $F_{ij}(a) = \Phi((a - \delta_i - \gamma_j)\sqrt{\lambda})$, Φ and ϕ are the standard normal CDF and density, and $c_\rho(u, v) = \partial^2 C_\rho / \partial u \partial v$ is the Frank copula density

$$c_\rho(u, v) = \frac{\rho(1 - e^{-\rho}) e^{-\rho(u+v)}}{[1 - e^{-\rho} - (1 - e^{-\rho u})(1 - e^{-\rho v})]^2}, \quad \rho \in \mathbb{R}, \quad (5)$$

The log-likelihood contribution of dyad $\{i, j\}$ is therefore

$$\log p(A_{ij}, A_{ji} \mid \theta, \rho) = \log c_\rho(\Phi(z_{ij}), \Phi(z_{ji})) + \log \phi(z_{ij}) + \log \phi(z_{ji}) + \log \lambda, \quad (6)$$

where $z_{ij} = (A_{ij} - \delta_i - \gamma_j)\sqrt{\lambda}$.

In particular, ρ is the only parameter governing within-dyad dependence; all remaining parameters affect only the marginal distributions. The dependence parameter $\rho \in \mathbb{R}$ governs *reciprocity*: when $\rho > 0$ the edges A_{ij} and A_{ji} are positively associated; when $\rho < 0$ they are negatively associated, implying a hierarchical or asymmetric regulation structure; and

141 $\rho = 0$ corresponds to independence (the Frank copula reduces to the independence copula in
 142 the limit $\rho \rightarrow 0$). Figure 1 illustrates these three regimes in the uniform (PIT) space after
 143 marginal structure is removed. Section 4 formalises these properties.

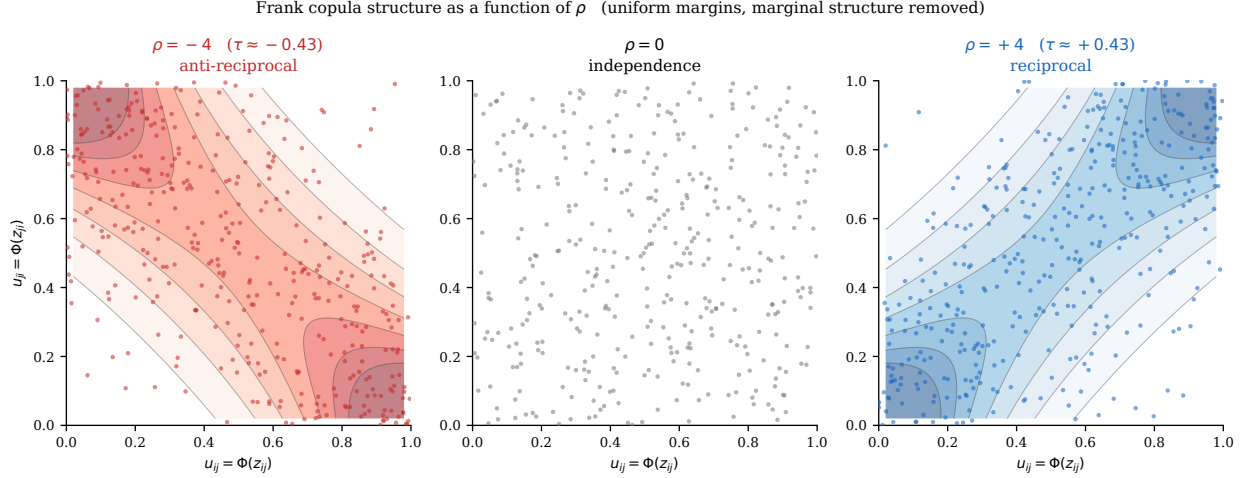


Figure 1: Frank copula structure in uniform (probability-integral-transform) space for three representative values of ρ . Each panel shows 400 simulated dyad pairs (u_{ij}, u_{ji}) with filled density contours overlaid. Left ($\rho = -4$): anti-diagonal concentration, meaning A_{ij} large $\Leftrightarrow A_{ji}$ small—hierarchical structure. Centre ($\rho = 0$): uniform, no dependence. Right ($\rho = +4$): diagonal concentration, meaning large A_{ij} co-occurs with large A_{ji} —reciprocal structure. The brain data correspond to the left panel.

144 The choice of copula family is guided by the empirical dependence structure of the data
 145 (Figure 2). The brain dyad pairs exhibit symmetric anti-diagonal concentration without
 146 evidence of tail asymmetry. This rules out asymmetric Archimedean families such as Clayton
 147 and Gumbel, which place excess mass in one tail corner of the unit square (Figure 2, right
 148 column). Among symmetric copulas, the Gaussian copula imposes an elliptical dependence
 149 structure that concentrates mass along linear contours; the Frank copula, by contrast, produces
 150 a sharper, non-elliptical anti-diagonal concentration that better matches the observed pattern
 151 (Figure 2, center versus left column).

152 The Frank family is therefore the simplest model that matches the observed symmetric anti-
 153 diagonal dependence structure. Beyond the empirical fit, it has three additional theoretical
 154 properties that make it natural here. First, it is the unique Archimedean family with radial

symmetry ($C_\rho(u, v) = C_\rho(v, u)$), appropriate for an undirected dyad where the roles of A_{ij} and A_{ji} are symmetric. Second, ρ is defined on all of $\mathbb{R}$, spanning the full range from perfect negative to perfect positive dependence with independence at $\rho = 0$, natural for reciprocity which can take either sign. Third, the one-to-one correspondence between ρ and Kendall's τ via the Debye function (Theorem 4.1) makes the parameter directly interpretable as a rank-correlation measure of reciprocity, invariant to marginal transformations. Sensitivity to copula choice is discussed in Section 9.

To quantify the visual comparison, Table 1 reports the mean per-dyad log copula density for each of the three families, evaluated on the 112 brain data participants using posterior mean marginal parameters. Copula parameters are matched to the same Kendall's τ via the standard moment-matching formulas ($r = \sin(\pi\tau/2)$ for the Gaussian copula and $\theta = -2\tau/(1 + \tau)$ for the 90°-rotated Clayton copula). Frank provides the highest mean log density (0.158), followed by the Gaussian copula (0.107), with Clayton substantially lower (-0.020) and with considerably higher variance across participants. The Frank copula is therefore the best-fitting symmetric copula on this dataset, both visually and in terms of predictive likelihood.

Table 1: Mean per-dyad log copula density for three families evaluated on the brain data using posterior mean marginal parameters. Copula parameters are matched to the same Kendall's $\tau \approx -0.43$ (Frank: posterior mean $\hat{\rho}$; Gaussian and Clayton: τ -matched). Frank provides the best average fit across all 112 participants.

Copula family	Mean	Median	SD
Frank	0.158	0.163	0.143
Gaussian copula	0.107	0.141	0.174
Clayton (90°)	-0.020	0.066	0.335

3.4 Bayesian Inference

We adopt a Bayesian framework with the following hierarchical prior:

$$\rho_0, \delta_0, \gamma_0 \sim \mathcal{N}(0, 4), \quad (7)$$

$$\rho^{(g)} \mid \rho_0, \sigma_\rho \sim \mathcal{N}(\rho_0, \sigma_\rho^2), \quad (8)$$

$$\delta_i \mid \delta_0, \sigma_\delta \sim \mathcal{N}(\delta_0, \sigma_\delta^2), \quad i = 2, \dots, N, \quad (9)$$

$$\gamma_j \mid \gamma_0, \sigma_\gamma \sim \mathcal{N}(\gamma_0, \sigma_\gamma^2), \quad j = 1, \dots, N, \quad (10)$$

$$\lambda \sim \text{Gamma}(5, 4), \quad (11)$$

$$\sigma_\rho, \sigma_\delta, \sigma_\gamma \sim \text{Gamma}(0.1, 10). \quad (12)$$

The hyper-means $(\rho_0, \delta_0, \gamma_0)$ pool information across groups and nodes respectively; the hyper-variances control the degree of shrinkage. Posterior samples are drawn via a componentwise Metropolis–Hastings algorithm, cycling through all parameters in sequence.

MAP versus full posterior sampling. Maximum a posteriori (MAP) estimation with a Laplace approximation is a natural fast alternative and provides useful diagnostics: the MAP point estimate is quick to compute, easy to cross-validate, and sufficient for the Kendall’s τ interpretation. We use full MCMC rather than MAP for three reasons specific to this application. First, ρ is the primary inferential target and its posterior can be non-symmetric at moderate $|\rho|$, making a diagonal Hessian approximation unreliable. Second, the hierarchical prior induces posterior dependence among group-level parameters that a Laplace approximation around the MAP does not capture. Third, full MCMC gives ranking probabilities directly: the posterior probability $\Pr(\rho^{(g_1)} < \rho^{(g_2)} < \rho^{(g_3)} \mid \text{data})$ is the most honest summary of the ordering claim and requires no additional calculation beyond counting posterior draws. We note that MAP estimates are consistent with the posterior means reported in Section 8, supporting approximate unimodality.

4 Theoretical Properties of the CGRG

This section derives several properties that clarify the interpretation and estimability of ρ . The main results establish its margin-invariant rank interpretation (Theorem 4.1), identifiability for $N \geq 3$ (Theorem 4.2), and $\sqrt{S}$ -consistency of the MLE under a simplified parametric model (Theorem 4.3). An entropy decomposition is deferred to Appendix A.

4.1 Reciprocity and Rank Correlation

Theorem 4.1 (Kendall's τ and Margin-Invariant Reciprocity). *Let (A_{ij}, A_{ji}) follow a CGRG with Frank copula parameter ρ and any continuous marginals. Define the first-order Debye function $D_1(\rho) = \rho^{-1} \int_0^\rho t(e^t - 1)^{-1} dt$, extended by continuity to $D_1(0) = 1$. Then:*

- (a) *Kendall's τ satisfies $\tau(\rho) = 1 - 4\rho^{-1}[1 - D_1(\rho)]$.*
- (b) *$\tau(\rho)$ is strictly increasing in ρ on $\mathbb{R}$, with $\tau(0) = 0$, $\tau(\rho) \rightarrow +1$ as $\rho \rightarrow +\infty$, and $\tau(\rho) \rightarrow -1$ as $\rho \rightarrow -\infty$.*
- (c) *$\text{sign}(\tau(\rho)) = \text{sign}(\rho)$.*
- (d) *$\tau(\rho)$ depends only on ρ , not on δ_i, γ_j , or λ . Therefore ρ provides a margin-invariant parametrization of reciprocity: its rank-correlation interpretation is unchanged by strictly increasing transformations of the marginals.*

Proof. Part (a). For any copula C , Kendall's τ satisfies $\tau = 4 \mathbb{E}_{(U,V) \sim C}[C(U,V)] - 1$ [3, Theorem 5.1.3]. Evaluating the expectation for the Frank family yields part (a); see Joe [4, Section 5.3].

Part (b). Differentiating $\tau(\rho)$ and using $D'_1(\rho) = \rho^{-1}[\rho(e^\rho - 1)^{-1} - D_1(\rho)]$ gives

$$\tau'(\rho) = \frac{4}{\rho^2} \left[1 - \frac{\rho}{e^\rho - 1} \right].$$

Since $e^\rho - 1 > \rho$ for $\rho > 0$ and $e^\rho - 1 < \rho$ for $\rho < 0$ (both by strict convexity of e^ρ), we have $\tau'(\rho) > 0$ for all $\rho \neq 0$. The limiting values follow from $D_1(\rho) \rightarrow 0$ as $\rho \rightarrow +\infty$ and $D_1(\rho) \rightarrow 1$ as $\rho \rightarrow 0$.

Part (c). Follows from $\tau(0) = 0$ and strict monotonicity.

Part (d). Kendall's τ is invariant under strictly increasing marginal transformations [3, Theorem 5.1.9]. Since normal CDFs are strictly increasing, the rank correlation depends only on C_ρ . $\square$

4.2 Identifiability

Theorem 4.2 applies to the simplified model of this section: Gaussian marginals with fixed parameters $(\delta_i, \gamma_j, \lambda)$, dyadic Frank copula with parameter ρ , and $S \geq 1$ i.i.d. replicated networks. Formal identifiability of the full hierarchical model in Section 5 involves additional considerations (shared hyperparameters and group-specific precision parameters) that are not addressed here; extending the argument to that setting is of theoretical interest and left for future work.

Theorem 4.2 (Identifiability). *Under the constraint $\delta_1 = 0$, the parameter vector*

$$\theta = (\delta_2, \dots, \delta_N, \gamma_1, \dots, \gamma_N, \lambda, \rho)$$

is identifiable from the joint distribution of $\{(A_{ij}, A_{ji}) : 1 \leq i < j \leq N\}$ whenever $N \geq 3$.

Proof. Denote the marginal mean $\mu_{ij} = \mathbb{E}[A_{ij}] = \delta_i + \gamma_j$ and the common marginal variance $\sigma^2 = \lambda^{-1}$.

Step 1: λ . Every marginal is $\mathcal{N}(\mu_{ij}, \lambda^{-1})$, so $\lambda^{-1} = \text{Var}(A_{ij})$ for any $i \neq j$. Hence λ is identified.

Step 2: Receiver effects γ_j for $j \geq 2$. Because $\delta_1 = 0$, we have $\mu_{1j} = \gamma_j$ for all $j \geq 2$. Thus $\gamma_j = \mu_{1j}$ is identified directly from the marginal mean of A_{1j} .

229 *Step 3: Sender effects δ_i for $i \geq 2$ and γ_1 .* For $i \geq 2$, we have $\mu_{i1} = \delta_i + \gamma_1$, so once γ_1 is
 230 known, $\delta_i = \mu_{i1} - \gamma_1$. It remains to identify γ_1 . For any $i \geq 2$ and $j \geq 2$ with $j \neq 1$,

$$\delta_i = \mu_{ij} - \gamma_j = \mu_{ij} - \mu_{1j},$$

231 using the expressions from Step 2. Substituting back into $\gamma_1 = \mu_{i1} - \delta_i$ gives

$$\gamma_1 = \mu_{i1} - (\mu_{ij} - \mu_{1j}).$$

232 This requires at least one pair (i, j) with $i, j \geq 2$ and $i \neq j$, which exists whenever $N \geq 3$.
 233 (With $N = 2$ the only pair is $(2, 1)$, giving $\delta_2 + \gamma_1 = \mu_{21}$, a single equation in two unknowns.)

234 *Step 4: ρ from the copula.* With λ and the mean structure identified, the marginal
 235 CDFs $F_{ij}(a) = \Phi((a - \mu_{ij})\sqrt{\lambda})$ are fully known. The joint CDF $H_{ij}(a, b) = C_\rho(F_{ij}(a), F_{ji}(b))$
 236 determines C_ρ via $C_\rho(u, v) = H_{ij}(F_{ij}^{-1}(u), F_{ji}^{-1}(v))$. By Sklar's theorem, the copula of a
 237 bivariate distribution with continuous marginals is unique. Since $\rho \mapsto C_\rho$ is injective for the
 238 Frank family (strict monotonicity of $\tau(\rho)$ in Theorem 4.1(b) implies $C_\rho \neq C_{\rho'}$ for $\rho \neq \rho'$), ρ
 239 is uniquely determined. $\square$

240 4.3 Asymptotic Inference

241 The asymptotic result below applies to the simplified CGRG of this section (fixed N , shared ρ ,
 242 i.i.d. replicates), not to the full hierarchical model of Section 5. The natural growing-sample
 243 regime for replicated network data is fixed number of nodes N and growing number of
 244 replicated networks S .

245 **Theorem 4.3** ($\sqrt{S}$ -Consistency of the MLE for ρ). *Fix $N \geq 3$ and let $S \rightarrow \infty$. Assume:*
 246 *(i) the S networks are independent replicates from the CGRG model with true parameter*
 247 *$\rho_0 \in \mathbb{R}$; (ii) all δ_i and γ_j are known, so ρ is the only unknown; (iii) the marginal parameters*
 248 *are such that the probability-integral-transformed dyad values (U_s^{ij}, V_s^{ij}) are i.i.d. c_{ρ_0} for each*

pair $i < j$. Define the Fisher information for ρ from the $m = N(N - 1)/2$ dyads of a single replicate:

$$\mathcal{I}(\rho) = m \cdot \mathbb{E}_{c_\rho} \left[\left(\frac{\partial}{\partial \rho} \log c_\rho(U, V) \right)^2 \right].$$

Then $\mathcal{I}(\rho) \in (0, \infty)$ for all $\rho \neq 0$, and the maximum likelihood estimator $\hat{\rho}_S$ satisfies

$$\sqrt{S}(\hat{\rho}_S - \rho_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\rho_0)^{-1}) \quad \text{as } S \rightarrow \infty.$$

The independence case $\rho_0 = 0$ requires separate local analysis and is not covered by this result.

Proof. The Frank log-copula density is smooth in ρ on $(0, 1)^2 \times (\mathbb{R} \setminus \{0\})$, with bounded score, giving $\mathcal{I}(\rho) < \infty$. Positivity follows from strict monotonicity of $\tau(\rho)$ (Theorem 4.1(b)). Asymptotic normality then follows from the standard M-estimator CLT [29, Theorem 5.41] applied to the i.i.d. sum $\mathcal{L}_S(\rho) = \sum_{s=1}^S \sum_{i < j} \ell_\rho(U_s^{ij}, V_s^{ij})$. $\square$

Remark 4.1. Theorem 4.3 fixes the marginal parameters and treats ρ as the sole unknown; it does not cover the full hierarchical model or joint estimation. Its value is to establish the $\sqrt{S}$ information-scaling for replicated-network data. Joint asymptotics under simultaneous estimation would require additional profile likelihood analysis and is left for future work.

5 Hierarchical CGRG for Replicated Brain Networks

5.1 Data and Network Construction

We apply the model to $S = 112$ resting-state fMRI-derived effective connectivity networks from Nie et al. [16]. Each network is a 4×4 matrix $A^{(s)}$ of connectivity strengths estimated by spectral dynamic causal modelling [15] for the four canonical nodes of the default mode network (DMN): medial prefrontal cortex (mPFC, node 1), posterior cingulate cortex (PCC, node 2), left inferior parietal lobule (LIPL, node 3), and right inferior parietal lobule (RIPL, node 4). These four regions are the core hubs of the DMN and are known to show altered

effective connectivity in Alzheimer’s disease [18–20]. Participants are classified as Alzheimer’s disease (AD, $S_{\text{AD}} = 12$), mild cognitive impairment (MCI, $S_{\text{MCI}} = 63$), or cognitively normal (NL, $S_{\text{NL}} = 37$) based on ADNI diagnostic criteria [13], with group labels $g_s \in \{\text{AD}, \text{MCI}, \text{NL}\}$. Throughout this section $N = 4$ denotes the number of DMN regions (nodes) and $S = 112$ denotes the total number of participants (replicated networks).

5.2 Multi-Group CGRG Model

The network edges fall into two types. *Self-connections* $A_{ii}^{(s)}$ capture the intrinsic decay of each region. *Off-diagonal edges* $A_{ij}^{(s)}$ ($i \neq j$) represent directed regulatory strengths and are the primary object of interest.

Following Section 3, the marginal distribution of each off-diagonal edge is

$$A_{ij}^{(s)} \mid \delta_i^{(g_s)}, \gamma_j^{(g_s)}, \lambda^{(g_s)} \sim \mathcal{N}\left(\delta_i^{(g_s)} + \gamma_j^{(g_s)}, (\lambda^{(g_s)})^{-1}\right), \quad (13)$$

where $\delta_i^{(g)}$ is the sender effect of node i in group g , measuring the tendency of region i to exert regulatory influence, and $\gamma_j^{(g)}$ is the receiver effect of node j , measuring the tendency of region j to be regulated. The precision $\lambda^{(g)}$ is group-specific. We set $\delta_1^{(g)} = 0$ for all groups. Self-connections are modeled marginally as $A_{ii}^{(s)} \sim \mathcal{N}(\delta_i^{(g_s)} + \gamma_i^{(g_s)}, (\lambda^{(g_s)})^{-1})$; they do not enter the copula.

Within each group g , the two directed edges of dyad $\{i, j\}$ are joined by a Frank copula with group-specific parameter $\rho^{(g)}$:

$$p(A_{ij}^{(s)}, A_{ji}^{(s)} \mid \rho^{(g_s)}, \dots) = c_{\rho^{(g_s)}}\left(F_{ij}^{(g_s)}(A_{ij}^{(s)}), F_{ji}^{(g_s)}(A_{ji}^{(s)})\right) \cdot f_{ij}^{(g_s)}(A_{ij}^{(s)}) \cdot f_{ji}^{(g_s)}(A_{ji}^{(s)}),$$

where $F_{ij}^{(g)}$ and $f_{ij}^{(g)}$ are the CDF and density of (13). The group log-likelihood is

$$\log p(\{A^{(s)}\}_{g_s=g} \mid \theta^{(g)}, \rho^{(g)}) = \sum_{s: g_s=g} \left[\sum_{i=1}^4 \log p(A_{ii}^{(s)}) + \sum_{i < j} \log p(A_{ij}^{(s)}, A_{ji}^{(s)}) \right]. \quad (14)$$

5.3 Hierarchical Prior

The sender effects, receiver effects, and copula parameters are linked across groups via shared hyperparameters, reflecting the belief that the functional roles of the four DMN nodes are similar across disease states:

$$\rho_0, \delta_0, \gamma_0 \sim \mathcal{N}(0, 4), \quad (15)$$

$$\rho^{(g)} \mid \rho_0, \sigma_\rho \sim \mathcal{N}(\rho_0, \sigma_\rho^2), \quad g = 1, 2, 3, \quad (16)$$

$$\delta_i^{(g)} \mid \delta_0, \sigma_\delta \sim \mathcal{N}(\delta_0, \sigma_\delta^2), \quad i = 2, 3, 4, \quad g = 1, 2, 3, \quad (17)$$

$$\gamma_j^{(g)} \mid \gamma_0, \sigma_\gamma \sim \mathcal{N}(\gamma_0, \sigma_\gamma^2), \quad j = 1, \dots, 4, \quad g = 1, 2, 3, \quad (18)$$

$$\lambda^{(g)} \sim \text{Gamma}(5, 4), \quad g = 1, 2, 3, \quad (19)$$

$$\sigma_\rho, \sigma_\delta, \sigma_\gamma \sim \text{Gamma}(0.1, 10). \quad (20)$$

The hyper-means $(\rho_0, \delta_0, \gamma_0)$ pool information across the three groups; the hyper-standard-deviations $(\sigma_\rho, \sigma_\delta, \sigma_\gamma)$ control the degree of shrinkage.

5.4 Posterior Computation

Posterior samples are drawn via a componentwise random-walk Metropolis–Hastings algorithm [16], cycling through ρ_0 ; $\rho^{(g)}$ for $g = 1, 2, 3$; δ_0 ; $\delta_i^{(g)}$ for $i = 2, 3, 4$ and $g = 1, 2, 3$; γ_0 ; $\gamma_j^{(g)}$ for $j = 1, \dots, 4$ and $g = 1, 2, 3$; $\lambda^{(g)}$ for $g = 1, 2, 3$; and $\sigma_\rho, \sigma_\delta, \sigma_\gamma$. Symmetric uniform proposals are used throughout. Burn-in of 10,000 iterations is discarded and every 100th draw of the subsequent 100,000 iterations is retained, yielding 1,000 approximately independent posterior draws. Step sizes were tuned during a pilot run to achieve acceptance rates in the 20–40% range for each block. The posterior probability of the strict group ordering, $\Pr(\rho^{(\text{NL})} < \rho^{(\text{MCI})} < \rho^{(\text{AD})} \mid \text{data}) = 0.20$, is computed directly by counting the fraction of retained draws satisfying the inequality; this single number summarises the confidence in the ordering claim without requiring any additional analytical approximation. Convergence is

assessed by running two independent chains from dispersed starting values and verifying that $\hat{R} < 1.05$ and the effective sample size exceeds 200 for each scalar parameter of interest.

6 Relationship to the Multivariate Normal

A natural alternative to the CGRG is a bivariate normal model for each dyad, with common marginal variance λ^{-1} and a dyad-specific correlation r_{ij} :

$$\begin{pmatrix} A_{ij} \\ A_{ji} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \delta_i + \gamma_j \\ \delta_j + \gamma_i \end{pmatrix}, \frac{1}{\lambda} \begin{pmatrix} 1 & r_{ij} \\ r_{ij} & 1 \end{pmatrix} \right).$$

The CGRG replaces this Gaussian joint distribution with a Frank-copula joint distribution sharing the same marginals. Three differences are worth noting.

First, the Frank copula is not a Gaussian copula, so the two models differ in their dependence structure even when Kendall's τ is matched via $r \approx \sin(\pi\tau/2)$. The Frank copula has symmetric, lighter tails than the Gaussian; for effective connectivity data, this means the CGRG does not force a specific tail dependence pattern that may not hold.

Second, the CGRG uses a single scalar ρ across all dyads within a group, whereas the dyad-specific bivariate normal uses $N(N-1)/2$ correlation parameters per group. This parsimony is the primary practical advantage in the replicated-network setting: with small S_g , a single ρ is estimable where dyad-specific correlations would require pooling assumptions or regularization.

Third, the margin-invariance of ρ (Theorem 4.1(d)) has no counterpart in the bivariate normal: Pearson correlation depends on the marginal distributions, whereas Kendall's τ and the Frank copula parameter do not. This makes ρ directly comparable across groups even when sender/receiver effects differ.

7 Simulation Study

We assess parameter recovery under two scenarios, each with $N = 5$ nodes and $S = 100$ replicated networks per group.

- **Scenario 1:** $\rho^{(1)} = -7$ (strong negative reciprocity), $\delta^{(1)} = (0, 2, -2, 0.25, -0.5)$, $\gamma^{(1)} = (15, 0.5, -2, 2, -1)$, $\lambda^{(1)} = 1$.

- **Scenario 2:** $\rho^{(2)} = -3$ (moderate negative reciprocity), $\delta^{(2)} = (0, 0.33, 8, -5, 9)$, $\gamma^{(2)} = (3, -12, 0, 4, -3)$, $\lambda^{(2)} = 1$.

Data were generated using the Frank copula with Gaussian marginals via the conditional quantile method: for each replicate and each dyad $\{i, j\}$, a pair (u, v) was drawn from the Frank copula C_ρ by sampling $u \sim \text{Uniform}(0, 1)$ and $v = C_\rho^{-1}(v|u)$ via conditional inversion, then transforming $(A_{ij}, A_{ji}) = (F_{ij}^{-1}(u), F_{ji}^{-1}(v))$ using the Gaussian marginal quantile functions. Table 2 summarises posterior performance from a single simulated dataset per scenario; it illustrates parameter recovery under the model but does not constitute a repeated-simulation coverage study. The copula parameter ρ is recovered less precisely than γ (larger SD relative to the scale of the parameter), as expected: ρ is identified only through higher-order joint structure, while γ_j is identified from marginal means. Figures 3 and 4 display the marginal posteriors of ρ under the two scenarios.

Additional simulations with $\rho = 0$ ($N = 5$, $S = 100$) confirm that the posterior concentrates near zero, verifying that the model does not spuriously detect reciprocity when none is present. This null-signal scenario mirrors the brain application result. A systematic repeated-simulation study reporting interval coverage and RMSE across many datasets is warranted and left for future work.

Table 2: Posterior summary for key parameters under Scenarios 1 and 2 ($N = 5$ nodes, $S = 100$ replicates per group). Columns: posterior mean, bias (mean $-$ true), posterior SD, and 95% credible interval. All intervals contain the true value.

Parameter	True	Mean	Bias	SD	95% CI
<i>Scenario 1</i>					
$\rho^{(1)}$	-7.00	-6.68	0.32	0.39	$[-7.44, -5.95]$
$\gamma_1^{(1)}$	15.00	14.97	-0.03	0.05	$[14.87, 15.08]$
$\gamma_2^{(1)}$	0.50	0.49	-0.01	0.08	$[0.33, 0.65]$
$\gamma_3^{(1)}$	-2.00	-2.07	-0.07	0.08	$[-2.22, -1.90]$
$\gamma_4^{(1)}$	2.00	1.95	-0.05	0.08	$[1.80, 2.11]$
$\gamma_5^{(1)}$	-1.00	-1.03	-0.03	0.08	$[-1.19, -0.88]$
<i>Scenario 2</i>					
$\rho^{(2)}$	-3.00	-2.88	0.12	0.30	$[-3.48, -2.31]$
$\gamma_1^{(2)}$	3.00	2.97	-0.03	0.08	$[2.80, 3.13]$
$\gamma_2^{(2)}$	-12.00	-12.08	-0.08	0.09	$[-12.26, -11.90]$
$\gamma_3^{(2)}$	0.00	-0.09	-0.09	0.09	$[-0.28, 0.09]$
$\gamma_4^{(2)}$	4.00	3.89	-0.12	0.10	$[3.69, 4.07]$
$\gamma_5^{(2)}$	-3.00	-3.13	-0.13	0.10	$[-3.31, -2.94]$

8 Results

8.1 Reciprocity

The CGRG model provides a direct test of dyadic dependence via ρ . Hierarchical (non-reciprocal) organisation implies that each region specialises as either a net sender or a net receiver, and would be characterised by negative ρ in the Frank copula model (Theorem 4.1). We test for this structure below.

Figure 5 shows posterior marginals of $\rho^{(g)}$ for the three groups. Posterior means and 95% credible intervals are reported in Table 3. All three posteriors are tightly concentrated around $\rho \approx -3.75$, well below zero, indicating strong negative dyadic dependence: participants with a stronger effective connection A_{ij} tend to have a weaker reverse connection A_{ji} , consistent with a hierarchical (non-reciprocal) regulatory structure in the DMN (Theorem 4.1). The posterior

for ρ corresponds to a Kendall's τ of approximately -0.43 via $\tau(\rho) = 1 - 4\rho^{-1}[1 - D_1(\rho)]$, matching the sample Kendall's τ of -0.37 to -0.55 observed across dyads.

The AD group, with only 12 participants ($n_{\text{AD}} = 72$ dyad observations), shows the widest posterior ($[-4.34, -2.91]$), as expected from the smaller sample, but it is still clearly concentrated below zero. All three groups have nearly identical posterior means for ρ ; the posterior probability of the strict ordering $\Pr(\rho^{(\text{NL})} < \rho^{(\text{MCI})} < \rho^{(\text{AD})} \mid \text{data}) = 0.20$, close to the uniform chance level of $1/6 \approx 0.17$, indicating that disease state does not substantially alter the degree of hierarchical reciprocity.

Table 3: Posterior summary of the Frank copula reciprocity parameter $\rho^{(g)}$ for the three diagnostic groups. Columns: posterior mean, posterior SD, and 95% credible interval. The final row reports the posterior probability of the strict ordering across all three groups, computed from the retained MCMC draws.

Group	Mean	SD	95% CI
NL	-3.778	0.294	$[-4.379, -3.209]$
MCI	-3.769	0.274	$[-4.326, -3.235]$
AD	-3.721	0.340	$[-4.331, -2.939]$
$\Pr(\rho^{(\text{NL})} < \rho^{(\text{MCI})} < \rho^{(\text{AD})} \mid \text{data}) = 0.1986$			

8.2 Node Effects

Table 4 summarises the posterior means and 95% credible intervals for the receiver effects $\gamma_j^{(g)}$ and sender effects $\delta_i^{(g)}$ across all four DMN nodes and three groups. The receiver effect $\gamma_j^{(g)}$ measures the average regulatory input received by node j in group g ; the sender effect $\delta_i^{(g)}$ ($i \geq 2$) measures the excess regulatory output of node i relative to the mPFC reference ($\delta_1^{(g)} = 0$). Full posterior coefficient plots for receiver and sender effects are provided in Appendix B.

Figure 9 shows the posterior mean connectivity matrix $E[A_{ij}] = \hat{\delta}_i + \hat{\gamma}_j$ and the asymmetry matrix $E[A_{ij}] - E[A_{ji}]$ for each group. Figure 8 summarises the same information as a directed

graph: node size encodes the receiver effect γ_j , arrow width encodes $E[A_{ij}]$, and arrow colour encodes asymmetry. LIPL (node 3) and RIPL (node 4) have the largest positive receiver effects, consistent with their role as integration hubs in the DMN. The asymmetry matrices and network diagram both show that mPFC is the strongest net sender, while PCC, LIPL, and RIPL are net receivers. These patterns are nearly identical across the three diagnostic groups.

Table 4: Posterior means and 95% credible intervals for receiver effects $\gamma_j^{(g)}$ and sender effects $\delta_i^{(g)}$ ($i \geq 2$) across the three diagnostic groups. Node labels: mPFC (node 1), PCC (node 2), LIPL (node 3), RIPL (node 4).

Parameter	Node	NL	MCI	AD
<i>Receiver effects $\gamma_j^{(g)}$</i>				
γ_1	mPFC	0.11 [0.06, 0.16]	0.11 [0.06, 0.16]	0.11 [0.06, 0.18]
γ_2	PCC	0.08 [0.02, 0.14]	0.08 [0.02, 0.13]	0.08 [0.02, 0.14]
γ_3	LIPL	0.18 [0.12, 0.24]	0.18 [0.13, 0.24]	0.18 [0.12, 0.25]
γ_4	RIPL	0.14 [0.08, 0.19]	0.14 [0.09, 0.20]	0.15 [0.09, 0.22]
<i>Sender effects $\delta_i^{(g)}$ (relative to mPFC)</i>				
δ_2	PCC	-0.07 [-0.14, -0.01]	-0.07 [-0.13, -0.00]	-0.06 [-0.13, 0.02]
δ_3	LIPL	-0.06 [-0.11, 0.00]	-0.06 [-0.11, 0.00]	-0.05 [-0.11, 0.02]
δ_4	RIPL	-0.04 [-0.11, 0.02]	-0.04 [-0.10, 0.02]	-0.04 [-0.11, 0.02]

8.3 Predictive Validation

Figure 10 assesses how much the Frank copula improves predictive fit relative to an independence baseline ($\rho = 0$, identical marginals). For each participant s in group g we compute the average copula log-density across the six DMN dyads,

$$\text{gain}_s = \frac{1}{6} \sum_{i < j} \log c_{\hat{\rho}^{(g_s)}}(u_{ij}^{(s)}, u_{ji}^{(s)}),$$

where $(u_{ij}^{(s)}, u_{ji}^{(s)})$ are the probability-integral-transformed edges under posterior mean parameters. Under the independence copula $\log c_0 = 0$, so $\text{gain}_s > 0$ means the copula model

386 assigns higher density to the observed pair than independence does. Because $\hat{\rho} \approx -3.75$ and
 387 the data show anti-diagonal structure (u_{ij} high $\Leftrightarrow u_{ji}$ low), the Frank copula concentrates
 388 mass precisely in the anti-diagonal region occupied by the observations, yielding consistently
 389 positive gains across all 112 participants and all three groups.

390 9 Discussion

391 This paper has developed a dyadic Frank-copula model for directed real-valued networks,
 392 combining an additive sender-receiver mean structure with a single margin-invariant reciprocity
 393 parameter ρ . Section 4 derives that ρ admits a closed-form interpretation via Kendall's
 394 τ , decomposes the within-dyad entropy, is identifiable for $N \geq 3$, and is $\sqrt{S}$ -consistently
 395 estimable in a simplified parametric benchmark.

396 The brain connectivity application reveals a strong and consistent signal. Posterior
 397 estimates of ρ are concentrated around -3.75 for all three diagnostic groups, with 95%
 398 credible intervals entirely below zero. A Kendall's τ of approximately -0.43 implies that,
 399 within each dyad, participants with stronger forward connectivity A_{ij} systematically have
 400 weaker reverse connectivity A_{ji} . This is precisely the hierarchical (non-reciprocal) regulation
 401 signature that the model is designed to detect. The three groups share nearly the same ρ :
 402 the posterior probability of the strict ordering $\rho^{(\text{NL})} < \rho^{(\text{MCI})} < \rho^{(\text{AD})}$ is 0.20, close to chance,
 403 indicating that the degree of hierarchical organisation does not change substantially across
 404 disease states in this cohort. The lack of a group difference may reflect genuine invariance of
 405 the hierarchical DMN structure across Alzheimer's progression, or insufficient statistical power
 406 to resolve a small between-group shift in ρ with these sample sizes (particularly $S_{\text{AD}} = 12$).
 407 Node-level receiver and sender effects in Table 4 and Figure 9 show positive receiver effects
 408 across all groups, with LIPL and RIPL showing the largest positive values, consistent with
 409 the known role of parietal regions as network hubs [19]. Posterior predictive checks (Figure 6)
 410 confirm that the model reproduces the observed dyadic Kendall's τ values, and the copula

gain analysis (Figure 10) shows that the Frank copula model assigns substantially higher density to the observed dyad pairs than the independence baseline for all 112 participants.

Several extensions are worth pursuing. First, a proper repeated-simulation study reporting interval coverage and RMSE over many datasets would replace the single-dataset illustration of Section 7. Fourth, the dyadic partition can be replaced by larger blocks via vine copula constructions [24]. Fifth, longitudinal brain networks require a dynamic version; the spectral DCM framework of Nie et al. [16] is one starting point. Sixth, a joint asymptotics result would require additional profile likelihood analysis. Seventh, extending identifiability to the full hierarchical model is of theoretical interest.

External validation datasets. The current brain application is limited to 112 participants, with only 12 in the AD group. Several publicly available resting-state fMRI datasets offer promising validation targets. For disease-relevant replication, the Alzheimer’s Disease Neuroimaging Initiative (ADNI) and the Open Access Series of Imaging Studies (OASIS-3) both include larger AD and MCI cohorts. For method stress-testing, the Human Connectome Project (HCP) Young Adult dataset provides 1,206 healthy participants with high-quality effective connectivity derivable from preprocessed timeseries. The COBRE dataset (72 schizophrenia patients, 75 controls) offers a convenient case-control replication in a different clinical domain. Validation on one of these datasets would substantially strengthen confidence in the model’s generalizability.

Relationship to deep learning methods. Graph neural networks (GNNs) and related deep learning approaches have been applied to larger neuroimaging datasets for diagnostic classification [19]. The CGRG is not a competitor to GNNs for prediction tasks but targets a different goal: estimating an interpretable, low-dimensional dependence parameter for hypothesis testing in a replicated-network setting. With $N = 4$ nodes and $S = 112$ participants, the data are far below the scale at which GNN approaches are effective; a neural model would overfit and obscure the parameter of scientific interest. GNN-based approaches are

appropriate when the goal is classification with large node sets and many subjects, not when the goal is estimating a bivariate dependence structure interpretable via Kendall’s τ . Future work combining subject-level graph encoders with copula-based dependence parameters for larger ROI-level networks remains an interesting direction.

A Entropy Decomposition

Theorem A.1 (Entropy Decomposition). *Let $h_C(\rho) = -\int_0^1 \int_0^1 c_\rho(u, v) \log c_\rho(u, v) du dv$ denote the copula entropy of the Frank copula, with $h_C(0) = 0$. Then: (a) $h_C(\rho) \leq 0$ with equality iff $\rho = 0$, with $h_C(\rho) = -D_{\text{KL}}(c_\rho \| c_0)$; (b) $h_C(\rho)$ is decreasing in $|\rho|$; (c) $H(A_{ij}, A_{ji}) = H(A_{ij}) + H(A_{ji}) + h_C(\rho)$; and (d) $I(A_{ij}; A_{ji}) = -h_C(\rho) \geq 0$.*

The proof is standard: part (a) follows from KL non-negativity, parts (b)–(c) from the concordance ordering of the Frank family and the probability integral transform, and part (d) from the definition of mutual information.

Corollary A.1.1 (Group Entropy Differences Under Equal Marginal Precision). *Under the CGRG model with $\lambda^{(g)} = \lambda$ for all g , $H_{\text{net}}^{(g)} - H_{\text{net}}^{(g')} = \frac{N(N-1)}{2} [h_C(\rho^{(g)}) - h_C(\rho^{(g')})]$. The copula term isolates the contribution of dyadic dependence when marginal precision is shared across groups.*

In the brain application the posterior estimates $\rho^{(g)} \approx -3.75$ yield $h_C(\rho^{(g)}) < 0$, meaning dyadic dependence reduces joint entropy relative to independence. The group-specific precision differences $\lambda^{(g)}$ contribute additionally to between-group entropy variation, and the two contributions can be separated via this decomposition. Figure ?? illustrates both components from the posterior.

B Posterior Coefficient Plots for Node Effects

Figure 11 displays posterior means and 95% credible intervals for all receiver effects $\gamma_j^{(g)}$ and sender effects $\delta_i^{(g)}$ ($i \geq 2$, relative to the mPFC reference) across the three diagnostic groups (NL: filled circles, MCI: squares, AD: triangles).

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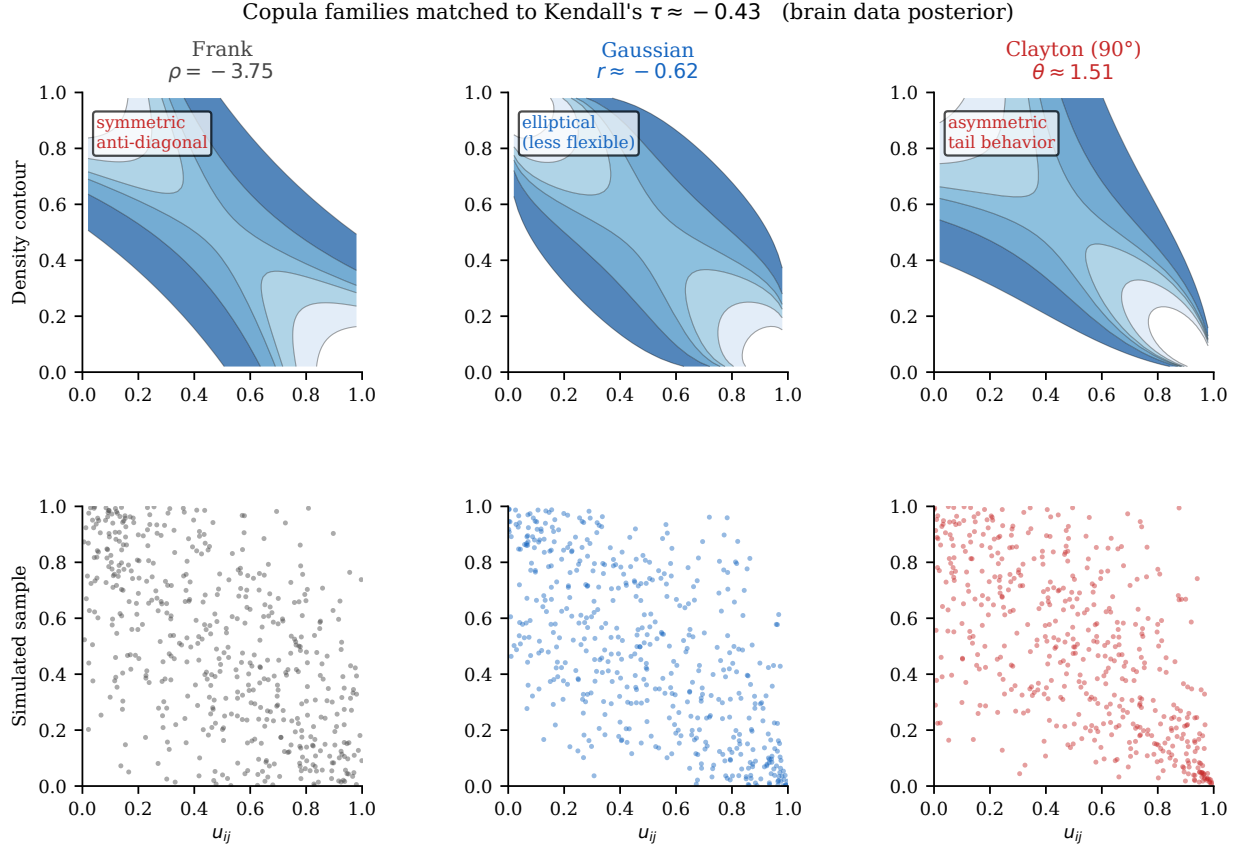


Figure 2: Comparison of three copula families at matched Kendall's $\tau \approx -0.43$ (the brain data posterior). Top row: density contours in uniform space. Bottom row: 500 simulated pairs. The Frank copula (left) produces symmetric anti-diagonal concentration that matches the observed dyad scatter (Figure 7). The Gaussian copula (centre) imposes an elliptical structure that concentrates mass along linear contours and provides a less sharp anti-diagonal signal. The 90°-rotated Clayton copula (right) produces asymmetric tail behaviour—with excess mass in the upper-left corner but not the lower-right—which is not supported by the data. The Frank family is the simplest model consistent with the symmetric anti-diagonal structure in the brain networks.

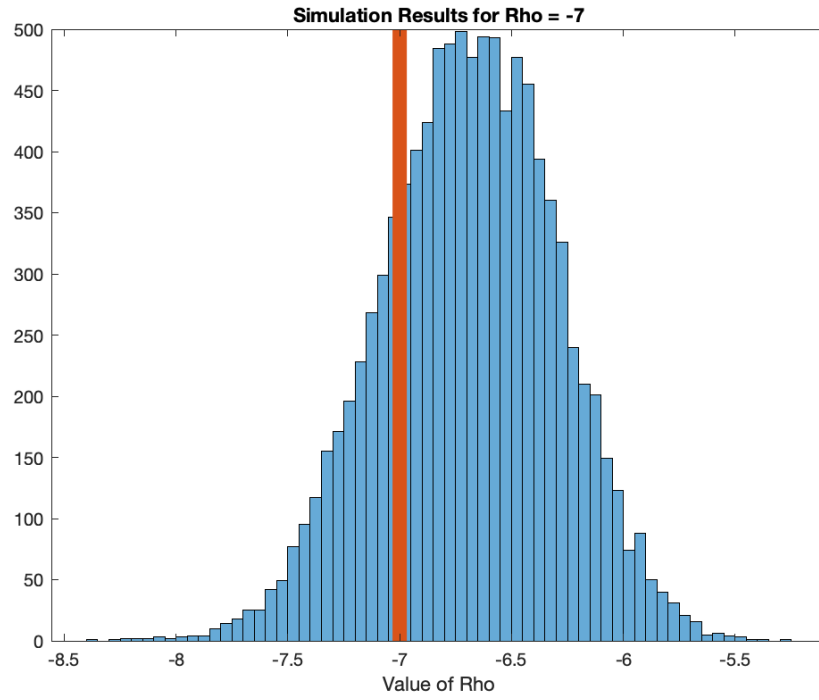


Figure 3: Simulation Scenario 1 ($\rho = -7$, $N = 5$, $S = 100$). Posterior marginal of ρ (solid line); true value shown as dashed vertical line. The posterior concentrates tightly near the true value.

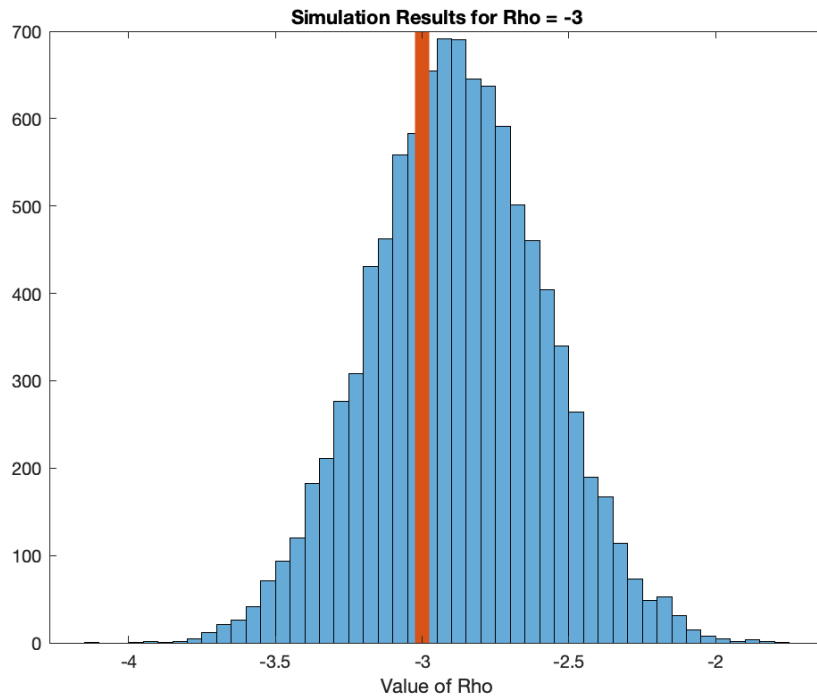


Figure 4: Simulation Scenario 2 ($\rho = -3$, $N = 5$, $S = 100$). Posterior marginal of ρ . The wider posterior relative to Scenario 1 reflects the lower signal-to-noise at moderate $|\rho|$.

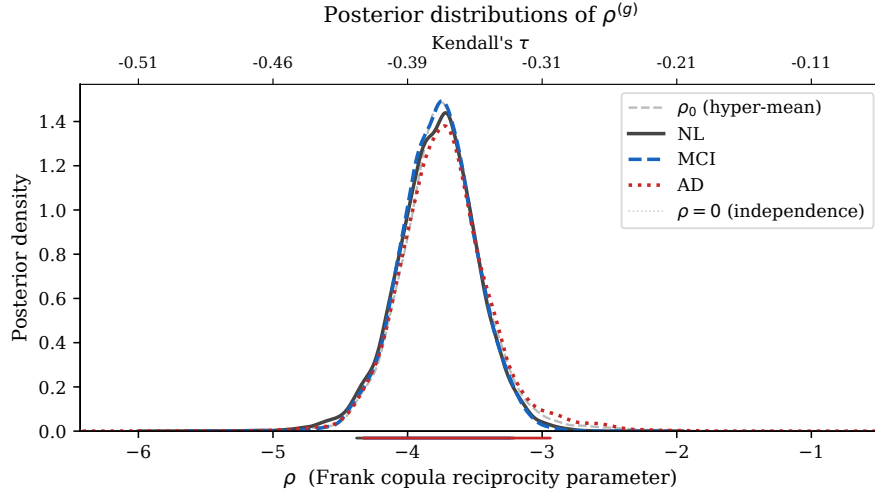


Figure 5: Posterior marginals of the Frank copula parameter $\rho^{(g)}$ for the three diagnostic groups (NL: grey, MCI: blue, AD: red). All three posteriors are concentrated around $\rho \approx -3.75$, well below zero, indicating strong negative dyadic dependence (hierarchical regulation structure) in the DMN. The upper x -axis shows the corresponding Kendall's τ via the Debye function. The AD group (12 participants) shows the widest posterior but is still clearly concentrated below zero.

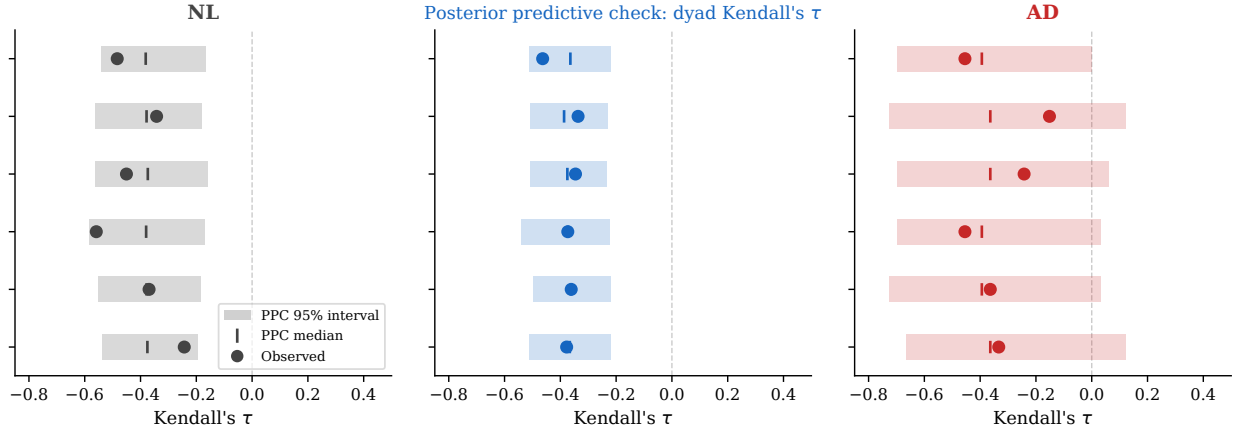


Figure 6: Posterior predictive check: dyadic Kendall's τ for the six DMN dyad pairs, by diagnostic group. Shaded bars show the 95% posterior predictive interval; vertical ticks show the posterior predictive median; filled circles show the observed sample Kendall's τ . The observed values fall within or close to the predictive intervals for all dyads, confirming that the fitted Frank copula reproduces the anti-diagonal dependence structure.

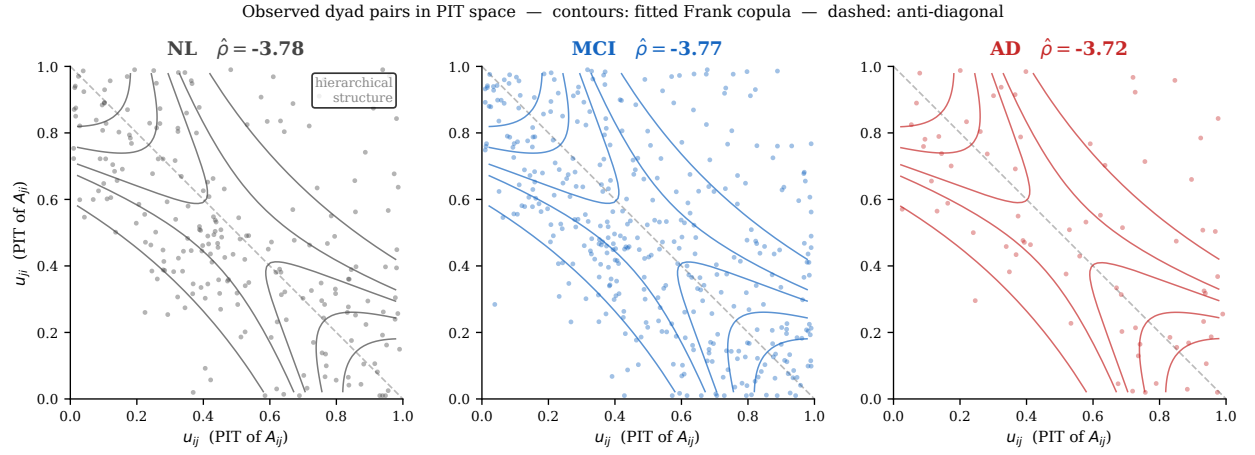


Figure 7: Observed dyad pairs (u_{ij}, u_{ji}) in probability-integral-transform (uniform) space after removing marginal structure, pooled across all six dyads and all subjects within each group. Contour lines show the fitted Frank copula density at the posterior mean $\hat{\rho}$ for each group. The anti-diagonal concentration of observations—and the close agreement between contours and data—confirms that the model captures the hierarchical (non-reciprocal) structure in the DMN. All three groups show the same qualitative pattern.

DMN posterior mean connectivity
(averaged over NL / MCI / AD — pattern identical)

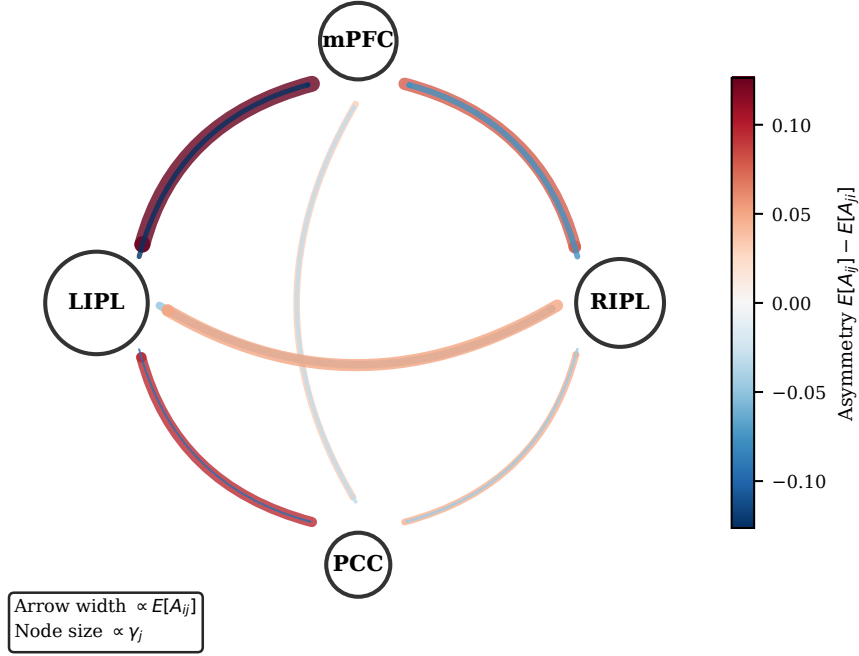


Figure 8: Directed-graph summary of posterior mean DMN connectivity (averaged over diagnostic groups; the pattern is nearly identical across NL, MCI, and AD). Node size is proportional to the receiver effect γ_j ; arrow width is proportional to $E[A_{ij}] = \hat{\delta}_i + \hat{\gamma}_j$; arrow colour shows asymmetry $E[A_{ij}] - E[A_{ji}]$ (red = forward direction dominant, blue = reverse direction dominant). mPFC is the dominant net sender; LIPL and RIPL are the dominant net receivers, consistent with their roles as parietal integration hubs.

Posterior mean connectivity and dyadic asymmetry across DMN nodes

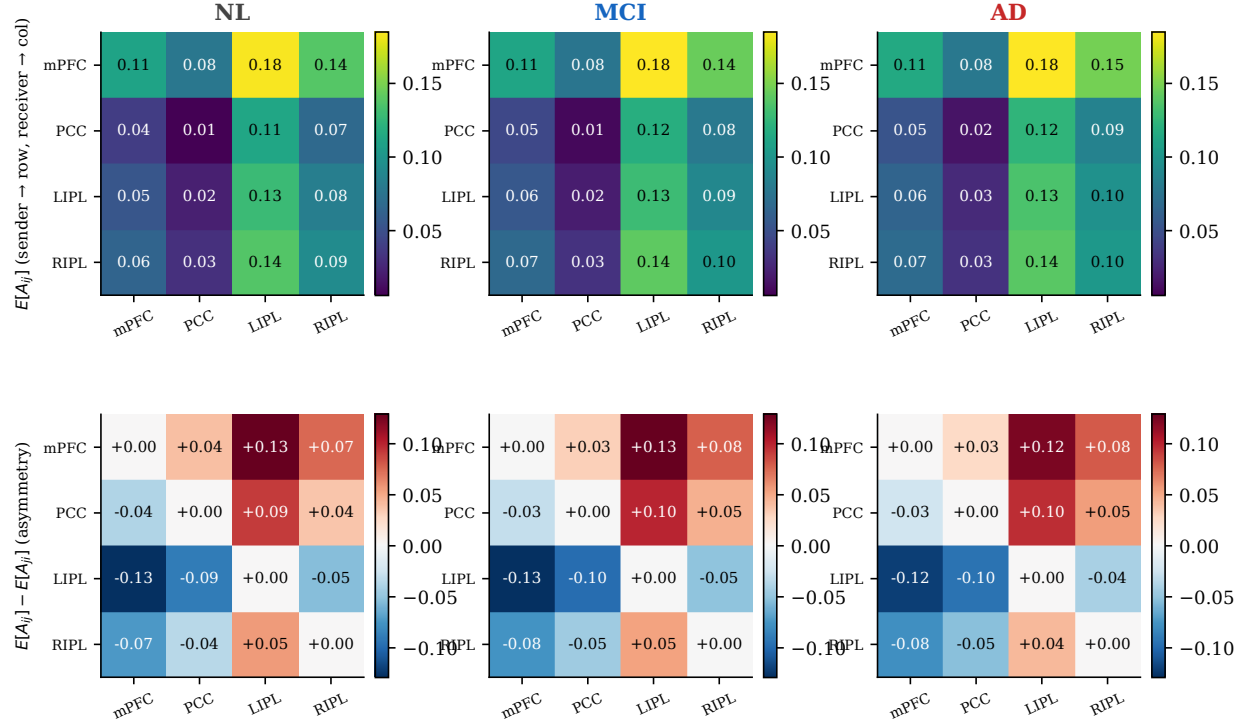


Figure 9: Top row: posterior mean connectivity $E[A_{ij}] = \hat{\delta}_i + \hat{\gamma}_j$ for each ordered pair of DMN nodes, by diagnostic group. Bottom row: asymmetry $E[A_{ij}] - E[A_{ji}]$ (red = node i sends more to j than j sends to i ; blue = reverse). LIPL and RIPL are the strongest net receivers; mPFC is the dominant net sender across all groups. The pattern is nearly identical across NL, MCI, and AD.

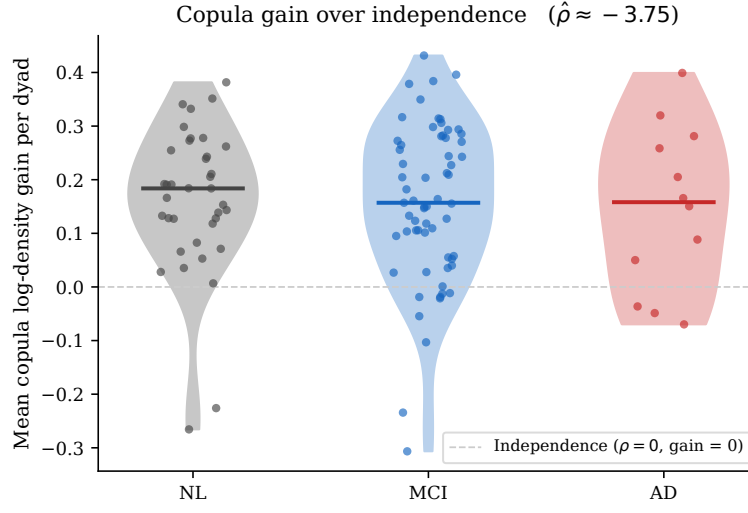


Figure 10: Copula log-density gain per dyad over the independence model ($\rho = 0$) for each participant, by diagnostic group. Positive values indicate that the fitted Frank copula assigns higher density to the observed (A_{ij}, A_{ji}) pair than independence. All 112 participants show positive gains; medians (horizontal bars) are near 0.35 nats per dyad. The dashed line marks the independence baseline.

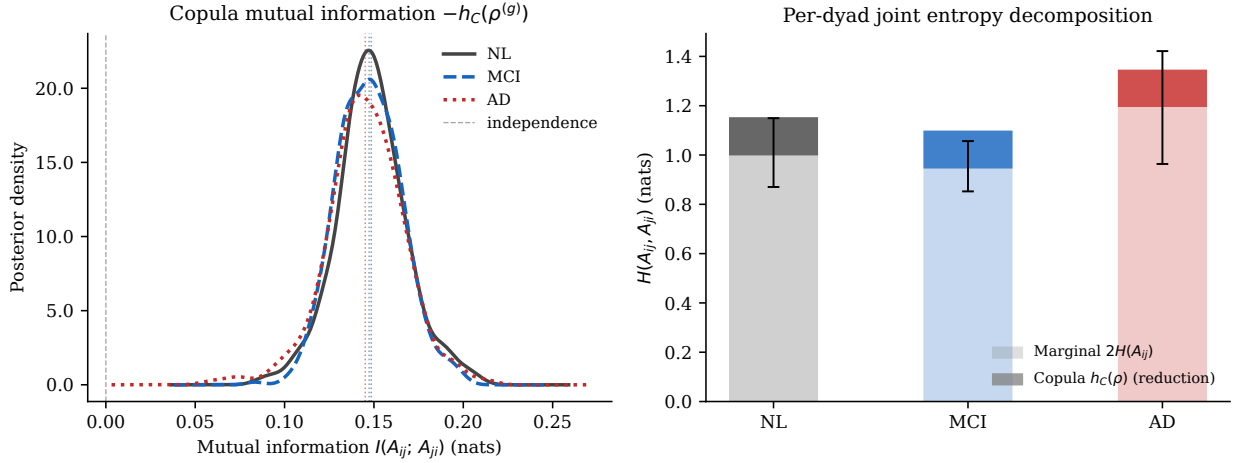


Figure 11: Entropy decomposition from the brain application posterior. *Left*: posterior distribution of mutual information $I(A_{ij}; A_{ji}) = -h_C(\rho^{(g)})$ for each diagnostic group. All three groups concentrate near 0.15 nats, confirming that within-dyad information is stable across disease stage. *Right*: per-dyad joint entropy $H(A_{ij}, A_{ji})$ decomposed into the marginal independence baseline $2H(A_{ij})$ (light bar) and the copula reduction $h_C(\rho^{(g)})$ (darker cap); error bars show 95% posterior credible intervals on the total. The AD group shows higher total entropy, driven by lower marginal precision $\lambda^{(AD)}$ rather than differences in ρ .

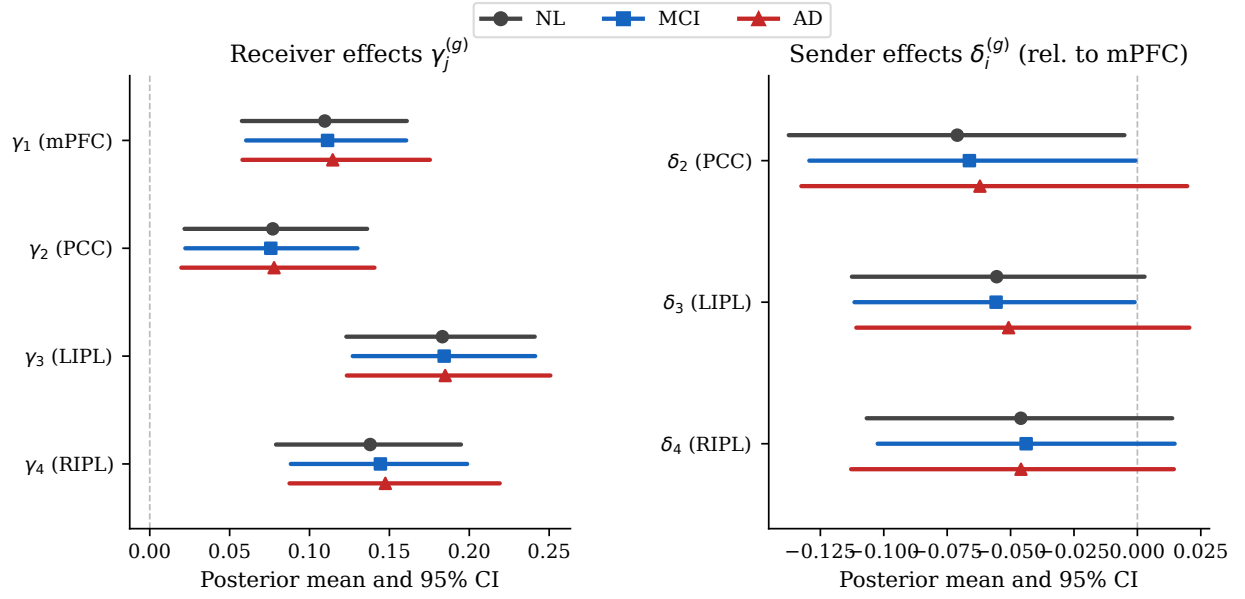


Figure 12: Forest plot of posterior means and 95% credible intervals for receiver effects $\gamma_j^{(g)}$ (left panel) and sender effects $\delta_i^{(g)}$ (right panel, relative to mPFC) for NL (dark grey circles), MCI (blue squares), and AD (red triangles). LIPL and RIPL show the largest positive receiver effects; PCC and LIPL show small negative sender effects relative to the mPFC reference across all groups. Credible intervals overlap substantially across groups, consistent with the finding that node-level effects are stable across diagnostic categories.